

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Contracting Activity: AFLCMC/PZIBA

Project/Program Name: HIGH SPEED COMBUSTION ANALYZER

Purchase Request / Local ID Number: [REDACTED]

Estimated Contract Cost (including Options): [REDACTED]

Section B: Description of the Supplies/Services Required

The Air Force Institute of Technology (AFIT) requires a single high-speed combustion powertrain analyzer to conduct crank-angle based diagnostics on small internal combustion engines. This unit is essential for capturing performance data for AFIT and multiple external agencies. The analyzer must be a 1U rack-mountable unit with at least 16 high-speed inputs (4 piezoelectric, 12 voltage), three CAN interfaces, and a +/- 10V input range. Key performance requirements include a minimum of 2GB of storage, 1000 Mbit/s Ethernet, a sampling rate of at least 2 million samples per second to resolve 200 kHz signals, and adjustable digital input filtering. Critically, the system must natively interface with the existing AVL IndiCom combustion analysis software, as the lab's established configuration files and software interfaces are dependent on it for processing high-speed combustion data.

Section C: Justification for Soliciting from a Single Source or Brand Name

This acquisition is restricted to the brand name AVL Test Systems, Inc. under the authority of FAR 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements, and FAR 8.405-6(a)(1)(i)(B), Limiting Sources, as only this brand-name item is compatible with the agency's existing equipment and essential to the government's requirements. AFIT has a substantial existing investment in AVL's proprietary "IndiCom" software, which includes extensive time, engineering hours, and training dedicated to developing custom configuration files and software interfaces. Procuring an analyzer from any other manufacturer would render this investment obsolete, necessitating a complete and costly reprogramming of the test infrastructure. Per FAR 6.302-1(a)(2)(ii), this would result in unacceptable delays of one year or more, severely disrupting ongoing research and collaborations. Therefore, to ensure continuity and prevent the duplication of costs that are not expected to be recovered through competition, the new analyzer must be compatible with the existing IndiCom software, making AVL the only viable source.

Section D: Efforts to Obtain Competition

Market research was conducted to ensure full and open competition to the maximum extent practicable. This research included internet searches, vendor outreach, and consultation with Subject Matter Experts at the Air Force Research Laboratory (AFRL), which confirmed that the mandatory compatibility with AFIT's existing AVL IndiCom software is the primary factor limiting competition.

Furthermore, to ensure all potential sources were identified, a Sources Sought notice was published using SAM.gov on 24 February 2026, with capability statements due on 16 March 2026 at 11:00 AM. As of the closing date and time, no responses or capability statements were received from industry. This lack of response further confirms that no other sources are available to meet the government's mandatory requirements.

The combination of initial market research and the absence of industry response to the Sources Sought notice solidifies the conclusion that no other source can satisfy the agency's needs. While other combustion analyzer manufacturers exist, none offer a solution that natively integrates with the required software, and AVL Test Systems, Inc. does not use authorized distributors or resellers for this specialized equipment.

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Section E: Steps to Preclude Future Single Source or Brand Name Awards

To foster competition in future acquisitions, the government will engage in continuous market research to identify new vendors and alternative technologies to the AVL IndiCom ecosystem. For all subsequent requirements, a thorough analysis will be performed to determine if the dependency on proprietary software can be reduced or eliminated, potentially by exploring platform-agnostic solutions or developing in-house software. The government also commits to ongoing industry engagement to communicate future needs and encourage the development of more competitive and interoperable solutions, thereby working to maximize competition wherever feasible in future procurements.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date
17-Mar-2026

